LAB 3 SE

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Architecture Selection:

We chose **Component-Based Architecture** for the Loan & Peer Credit Risk Assessment System.

1. Architectural Choice

The system uses a **Component-Based Architecture** in which the main functionality is divided into separate subsystems and components: **User Interface, Loan Processing, Credit Risk Assessment, and Database**. The components communicate through defined ports and interfaces. The architecture supports major functions such as loan application, credit risk assessment, loan funding, repayment, status tracking, and peer feedback review. These functions are derived from the system's functional requirements FR-01 to FR-06.

2. Two Reasons for the Choice

- **Modular functionality:** Loan application and evaluation, credit risk assessment and scoring, user interaction, and data storage are separated into independent components. This makes the system easier to develop, maintain, and modify.
- **Clear communication between modules:** The components exchange required information such as **loan application details, borrower data, risk scores, loan status, repayment information, and database information** through defined ports and interfaces. This provides clear communication between the different parts of the system.

3. Security Advantage

The architecture exposes functionality through **defined interfaces** instead of allowing every component to directly access another component's internal operations. For example, borrower financial information and lending data can be exchanged through the appropriate interfaces. This supports controlled communication and is consistent with the system's **Security & Privacy** requirement, which requires financial and lending data to be securely stored and transmitted and unauthorized access to be prevented.

4. Performance Benefit

Independent components can handle different operations without requiring the entire system to be processed as one unit. For example, **loan processing and credit-risk**

assessment can perform their respective operations when the required borrower and loan data is available. This separation can reduce unnecessary processing and supports the system's **Performance & Reliability** requirement for acceptable response times during concurrent transactions.

5. Component Diagram

